

Simulating High-Resolution 3D Smoke with 2D Turbulence Transfer

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1 INTRODUCTION

Visual simulation of fluid phenomena, such as smoke, using computer graphics has been used in various applications such as movies, video games, and commercial films. Creating realistic animation of high-resolution smoke with desired behaviors, however, is often time-consuming. The user has to adjust many physical-parameters via a trial-and-error process, which requires substantial effort and time. Sato et al. proposed a promising method to address this problem, called turbulence transfer, which transfers precomputed high-resolution details into a low-resolution simulation of smoke [Sato et al. 2018]. This method allows us to synthesize high-resolution smoke with the desired behavior. However, the method is still time-consuming; the 3D transfer process is computationally very expensive.

This poster presents a new turbulence transfer method to efficiently synthesize high resolution 3D smoke by using 2D turbulence transfer. Instead of the 3D transfer, we use 2D transfer for a set of cross section of the velocity distribution of the 3D smoke. In order to preserve the mass conservation condition, we convert the velocity distribution into a stream function. We investigate the quality and the performance of our method with different numbers of cross sections.

2 RELATED WORK

Several methods have been proposed for synthesizing high-resolution 3D smoke animation using 2D simulation. Rasmussen et al. use 2D simulation for synthesizing large-scale smoke [Rasmussen et al. 2003]. The method can simulate very high-resolution nuclear explosion. Although highly realistic animation can be generated, this method uses 2D simulation only and 3D motion of the smoke cannot be reproduced. Horvath and Geiger also use 2D simulation to

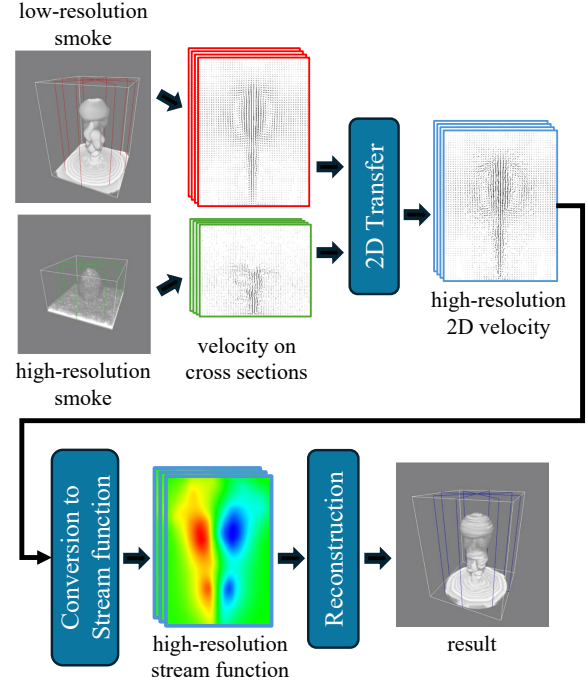


Figure 1: Overview of our method.

synthesize high-resolution fire [Horvath and Geiger 2009]. This method combines 3D particle simulation with 2D simulation on multiple planes arranged in perpendicular to the view direction. This method can take into account the 3D motion of fluids, but the camera motion is limited.

Our method can be considered as an extension of Rasmussen's method [Rasmussen et al. 2003], combined with the turbulence transfer method [Sato et al. 2018].

3 PROPOSED METHOD

Figure 1 illustrates an overview of our method. The input to our method includes a low-resolution 3D velocity field and a precomputed high-resolution 3D velocity field. We then generate multiple cross sections for both the input and the precomputed velocity fields as shown in Figure 1. Next we calculate a 2D velocity field on each cross sectional plane by projecting the 3D velocity onto the plane. The 2D velocity field is computed for both the low-resolution and the precomputed high-resolution velocity fields. We then apply

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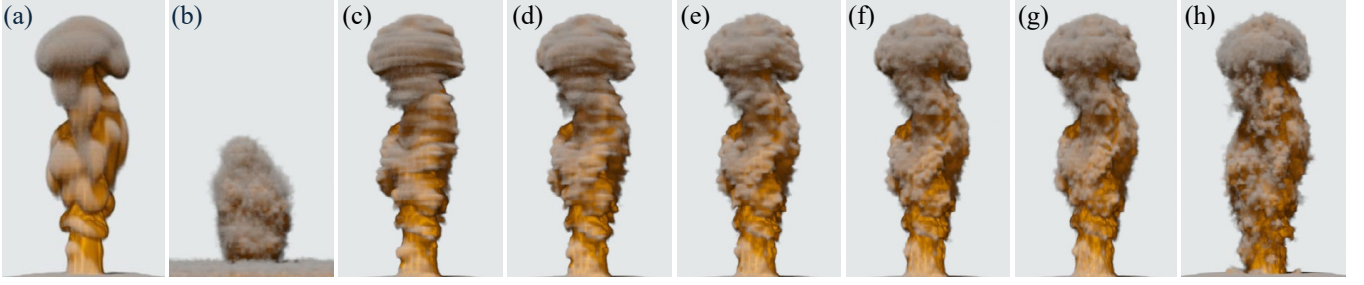


Figure 2: Experimental results. (a) input low-resolution smoke. (b) precomputed high-resolution smoke. (c) - (g) our results with different numbers of cross sections, $N_s = 4, 8, 16, 32, 64$, respectively. (h) result using 3D turbulence transfer.

the turbulence transfer method [Sato et al. 2018] to the 2D velocity field for each cross section. The high-frequency component of the transferred 2D velocity field is then transferred back to the low-resolution 3D velocity field.

Although the above method can synthesize high-resolution details efficiently, the resulting 3D velocity field does not preserve the mass conservation condition, that is, the velocity field is not a divergence-free field. In order to address this problem, we convert the 2D velocity field into a stream function after the turbulence transfer process as shown in Figure 1. The stream function in 2D is usually considered as a scalar function, denoted by $\phi(u_i, v_i)$, where (u_i, v_i) denotes a point with a local coordinate on i -th cross section. However, the stream function can also be considered as a function of a 3D vector, denoted as $\mathbf{q}(u_i, v_i, w_i) = (0, 0, \phi(u_i, v_i))$, where w_i corresponds to the normal vector of the cross section. We then generate a high-resolution 3D volume of the stream function by interpolating the stream functions on the cross sections. The 3D turbulent velocity field is then generated by applying the curl operator to the 3D volume, and the final high-resolution velocity field is obtained by adding the input low-resolution velocity field to it. Since the turbulent details are obtained from the stream function, the resulting velocity field is divergence-free by definition.

4 RESULTS

In this section, we present experimental results of smoke created using the proposed method. The computational environment used is a desktop PC with an Intel(R) Core(TM) i9-10980XE CPU (3.00GHz) with 128GB of memory. We use a resolution of $24 \times 24 \times 32$ for the low-resolution smoke and $192 \times 192 \times 128$ for the precomputed high-resolution smoke. Our method creates high resolution smoke with a resolution of $192 \times 192 \times 256$. Compared to the 3D turbulent transfer [Sato et al. 2018], which employs a patch size of $16 \times 16 \times 16$ resulting in 2304 patches, our proposed method uses a patch size of 16×16 , resulting in 192 patches per cross-section.

We render images using Maya, a 3D computer graphics software, based on the advected density field from the output flow. Figure 2 shows images created with different numbers of the cross sections, ranging from 4 to 64, comparing them with smoke generated using the 3D transfer method. From this figure, we observe that our method achieves results with visual accuracy comparable to those obtained by the 3D transfer method when the number of the cross sections is larger than 16.

When the number of the cross section is small, the linear interpolation between the cross sections significantly affects the results, leading to visible horizontal stripes. However, increasing the number reduces the impact of linear interpolation, resulting in plausible turbulent details. Table 1 presents the computation time taken for each computation. This table shows that increasing the number of the cross sections increases the computation time. Our method is faster when the number of the cross sections, N_s , is smaller than 32. The supplemental video shows the animation of the smoke with $N_s = 32$.

Table 1: Computation time in seconds for a single time step. Columns 2 through 6 show the computation times using our method with different number of cross sections N_s . The rightmost column shows the computation time using the 3D transfer method.

	2D transfer					3D transfer
N_s	4	8	16	32	64	
time	1.4	2.8	5.6	11.8	23.6	21.2

5 CONCLUSIONS

In this article, we have proposed the method for creating high-resolution smoke efficiently by 2D turbulent transfer. One of the future challenges is to enhance the versatility of the method. While our method focused on simulating smoke rising vertically without considering the influence of wind, in reality, smoke can be distorted along its central axis due to wind effects. Therefore, there is a need for a method that can handle such smoke scenarios.

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